

University of Barishal
Department of Computer Science and Engineering
Course Title: Object Oriented Programming
Course Code: CSE-2107
2nd Year 1st Semester Final Examination
Admission Session: 2019-2020

Time: 3 Hours

Marks: 60

N.B.: Answer any **FIVE** questions out of the followings. All parts of each question must be answered consecutively. Right side of the question shows the maximum marks.

- 1.a) Distinguish between the following terms and provide suitable examples for each: 8
- i) Objects and Classes
 - ii) Data abstraction and data encapsulation
 - iii) Inheritance and polymorphism
 - iv) Dynamic binding and message passing

- b) What will be output of the below code? Write answer with explanation. 4

i) <pre>public class Test { public static void main (String[]a) { if (-1) { break; } } }</pre>	ii) <pre>public class Test { public static void main(String[] args) { System.out.println("C" + "S" + "E"); } }</pre>
iii) <pre>class Test { public static void main(String[] as) { for(int i = 0; 1; i++) { System.out.println("Hello"); break; } } }</pre>	iv) <pre>public class CSE7 { public static void main(String[] arr){} public static void main (String arr){ } }</pre>

- 2.a) Draw the class diagram that shows inheritance, association, aggregation, composition relationship between classes for a real-world scenario for **Banking Application**. 12

- 3.a) Design a class to represent Bank Account. Include the following members: 6

Data members: • Name of the depositor • Account number • Type of account • Balance amount in the account	Methods: • To assign initial values • To deposit an amount • To withdraw an amount • To display (name and balance)
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N.B.: You do not need to write complete source code.

- b) What is inheritance and how does it help us create new classes quickly? Explain with example. 6
- 4.a) What is Constructor in Object Oriented Programming? Can it possible to write multiple constructors inside a class with different name? Explain your answer. 4
- b) Create a Java program that implements an abstract class Animal that has a Name property of type text and three methods SetName (string name), GetName and Eat. The Eat method will be an abstract method of type void. 5

You will also need to create a **Dog** class that implements the above **Animal** class and the **Eat** method that says the 'Dog is Eating'.

To create a new **Dog** type object from the Main of the program using the dog's name, give the Dog object a name, and then execute the **GetName** and **Eat** methods.

- c) Write down the difference between methods overwrite and overloading. Explain those with example. 3

5.a) What is JVM? Explain the internal architecture of JVM with neat sketch.

b) How does **String** class differ from the **StringBuffer** class?

c) An election is contested by 5 (five) candidates. The candidates are numbered 1 to 5 and the voting is done by making the candidate number on the ballot paper. Write a program to read the ballots and count the votes cast for each candidate using an array variable count. In case, a number read is outside the range 1 to 5, the ballot should be considered as 'spoilt ballot' and the program should also count the number of spoilt ballots.

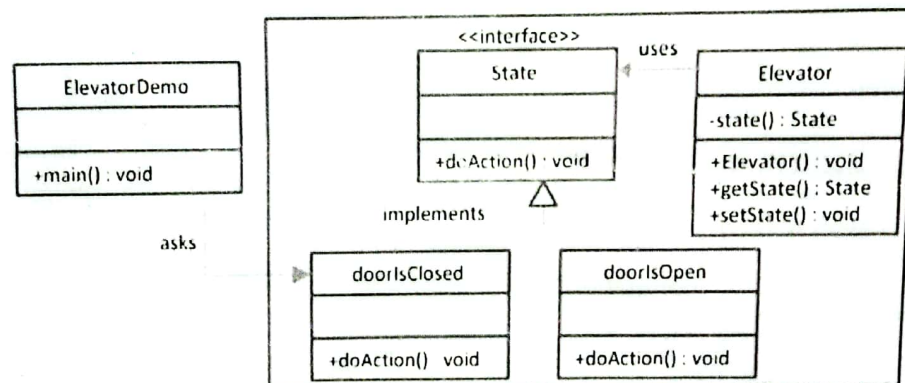
6.a) Describe the purpose of Java's wrapper classes, that is what they provide to programmers that their underlying values can't provide?

b) List some of the most common types of exceptions that might occur in Java. Give examples.

c) What is a Java Package and How is it used? Give example.

7.a) Threads can be used to make your Java applications faster. They allow you to run multiple things at the same time. You must think of a good scenario for using threads in real-world and discuss your scenario.

b) Consider an **Elevator Control Application** where the elevator is allowed to move only if the elevator doors are closed. This can be achieved using the State design pattern with two states doorIsOpen and doorIsClosed. The class diagram is show below.



Write the Java code for the `main()` method in the **ElevatorTest** class. You may assume that all the other classes are fully implemented. In the `main()` method of the **ElevatorTest** class, first create a object instance of **Elevator**, then make it execute `doAction()` when the door is open, and make it `doAction()` again when the door is closed.

8.a) The Collections class of Java Collection Framework has several methods (e.g. `sort()`, `min()`, `max()`) that require natural ordering of the classes to be handled. Explain the two possible conditions that must be satisfied on the classes to be able to use the `sort()` method to arrange the list of objects these classes.

b) Explain why you would choose a **LinkedHashSet** (or **LinkedHashMap**) over a **HashSet** (or **HashMap**, respectively).

c) What is a random-access file? How is it different from a sequential file? Why we need a random-access file?

"We cannot solve problems with the kind of thinking we employed when we came up with them."
- Albert Einstein